## ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 28 | $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ |
| 2 | **(a)** 30 | $\alpha\beta(\alpha+\beta)$ |
| 3 | **(a)** 7 | $\sqrt{(\alpha+\beta)^2-4\alpha\beta}$ |
| 4 | **(a)** $\frac{5}{2}$ | $\frac{\alpha^2+\beta^2}{\alpha\beta}$ |
| 5 | **(a)** 11 | $\frac{c}{a}$ for the monic cubic |
| 6 | **(c)** 21 | $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ |
| 7 | **(d)** 7 | Sum of zeroes $=9$, so other zero $=9-2$ |
| 8 | **(d)** 72 | $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ |
| 9 | **(b)** 56 | $\alpha\beta(\alpha+\beta)$ |
| 10 | **(a)** 7 | $\sqrt{(\alpha+\beta)^2-4\alpha\beta}$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | New sum $=\alpha^2+\beta^2=10$ (1); product $=(\alpha\beta)^2=9$; polynomial $x^2-10x+9$ (1). | 1+1 |
| 12 | Sum $=6$, product $=k$ (1); $k=6$ (1). | 1+1 |
| 13 | Zeroes $\alpha$ and $3\alpha$: $4\alpha=8$ so $\alpha=2$ (1); $k=3\alpha^2=12$ (1). | 1+1 |
| 14 | New sum $=\alpha^2+\beta^2=13$ (1); product $=(\alpha\beta)^2=36$; polynomial $x^2-13x+36$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\alpha+\beta=3$, $\alpha\beta=2$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{9}{2}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{9}{2}$ (1). | 1+1+1 |
| 16 | $(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $100-16=4c$ (1); $c=21$ (1). | 1+1+1 |
| 17 | $\sum\alpha=6$, $\sum\alpha\beta=11$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=14$ (1). | 1+1+1 |
| | **OR:** $\alpha+\beta=5$, $\alpha\beta=6$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{35}{6}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{49}{6}$ (1). | 1+1+1 |
| 18 | $(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $144-36=4c$ (1); $c=27$ (1). | 1+1+1 |
| | **OR:** $\sum\alpha=7$, $\sum\alpha\beta=14$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=21$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Let zeroes be $a-d, a, a+d$; sum $=3a=6$ so $a=2$ (2); product $=a(a^2-d^2)=6$ gives $d^2=1$ (2); zeroes $1, 2, 3$ (1). | 2+2+1 |
| | **OR:** $p(x)=g(x)q(x)+r(x)$ so $g(x)=\frac{p(x)-r(x)}{q(x)}$ (2); $p(x)-r(x)=g(x)q(x)=x^3+3x^2+3x+1$ (1); dividing, $g(x)=x+1$ (2). | 2+1+2 |
| 20 | Let zeroes be $a-d, a, a+d$; sum $=3a=9$ so $a=3$ (2); product $=a(a^2-d^2)=15$ gives $d^2=4$ (2); zeroes $1, 3, 5$ (1). | 2+2+1 |
| | **OR:** $p(x)=g(x)q(x)+r(x)$ so $g(x)=\frac{p(x)-r(x)}{q(x)}$ (2); $p(x)-r(x)=g(x)q(x)=x^3-2x^2+3x-6$ (1); dividing, $g(x)=x-2$ (2). | 2+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** $\frac{37}{6}$ | $\frac{\alpha^2+\beta^2}{\alpha\beta}$ |
| 2 | **(d)** 26 | $\frac{c}{a}$ for the monic cubic |
| 3 | **(a)** 35 | $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ |
| 4 | **(a)** 8 | Sum of zeroes $=11$, so other zero $=11-3$ |
| 5 | **(b)** 124 | $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ |
| 6 | **(c)** 120 | $\alpha\beta(\alpha+\beta)$ |
| 7 | **(d)** 7 | $\sqrt{(\alpha+\beta)^2-4\alpha\beta}$ |
| 8 | **(b)** $\frac{5}{2}$ | $\frac{\alpha^2+\beta^2}{\alpha\beta}$ |
| 9 | **(d)** 3 | $\frac{c}{a}$ for the monic cubic |
| 10 | **(b)** 40 | $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Sum $=9$, product $=k$ (1); $k=9$ (1). | 1+1 |
| 12 | Zeroes $\alpha$ and $4\alpha$: $5\alpha=10$ so $\alpha=2$ (1); $k=4\alpha^2=16$ (1). | 1+1 |
| 13 | New sum $=\alpha^2+\beta^2=26$ (1); product $=(\alpha\beta)^2=25$; polynomial $x^2-26x+25$ (1). | 1+1 |
| 14 | Sum $=11$, product $=k$ (1); $k=11$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\alpha+\beta=5$, $\alpha\beta=4$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{25}{4}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{25}{4}$ (1). | 1+1+1 |
| 16 | $(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $196-64=4c$ (1); $c=33$ (1). | 1+1+1 |
| 17 | $\sum\alpha=9$, $\sum\alpha\beta=26$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=29$ (1). | 1+1+1 |
| | **OR:** $\alpha+\beta=7$, $\alpha\beta=12$ (1); sum $=(\alpha+\beta)+\frac{\alpha+\beta}{\alpha\beta}=\frac{91}{12}$ (1); product $=\alpha\beta+2+\frac{1}{\alpha\beta}=\frac{169}{12}$ (1). | 1+1+1 |
| 18 | $(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$ (1); $256-100=4c$ (1); $c=39$ (1). | 1+1+1 |
| | **OR:** $\sum\alpha=7$, $\sum\alpha\beta=7$ (1+1); $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta=35$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Let zeroes be $a-d, a, a+d$; sum $=3a=12$ so $a=4$ (2); product $=a(a^2-d^2)=28$ gives $d^2=9$ (2); zeroes $1, 4, 7$ (1). | 2+2+1 |
| | **OR:** $p(x)=g(x)q(x)+r(x)$ so $g(x)=\frac{p(x)-r(x)}{q(x)}$ (2); $p(x)-r(x)=g(x)q(x)=x^3-x^2+x-1$ (1); dividing, $g(x)=x^2+1$ (2). | 2+1+2 |
| 20 | Let zeroes be $a-d, a, a+d$; sum $=3a=3$ so $a=1$ (2); product $=a(a^2-d^2)=-3$ gives $d^2=4$ (2); zeroes $-1, 1, 3$ (1). | 2+2+1 |
| | **OR:** $p(x)=g(x)q(x)+r(x)$ so $g(x)=\frac{p(x)-r(x)}{q(x)}$ (2); $p(x)-r(x)=g(x)q(x)=2x^2+7x+3$ (1); dividing, $g(x)=2x+1$ (2). | 2+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 10 | Sum of zeroes $= 13$, so other zero $= 13 - 3$ |
| 2 | **(d)** 133 | $(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$ |
| 3 | **(a)** -48 | $\alpha\beta(\alpha+\beta)$ |
| 4 | **(a)** 8 | $\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$ |
| 5 | **(b)** $\frac{29}{10}$ | $\dfrac{\alpha^2+\beta^2}{\alpha\beta}$ |
| 6 | **(b)** -8 | $\frac{c}{a}$ for the monic cubic |
| 7 | **(a)** 15 | $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$ |
| 8 | **(a)** 8 | Sum of zeroes $= 12$, so other zero $= 12 - 4$ |
| 9 | **(d)** 91 | $(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$ |
| 10 | **(d)** 180 | $\alpha\beta(\alpha+\beta)$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Zeroes $\alpha$ and $2\alpha$: $3\alpha = 12$ so $\alpha = 4$ (1); $k = 2\alpha^2 = 32$ (1). | 1+1 |
| 12 | New sum $= \alpha^2+\beta^2 = 20$ (1); product $= (\alpha\beta)^2 = 64$; polynomial $x^2 - 20x + 64$ (1). | 1+1 |
| 13 | Sum $= 13$, product $= k$ (1); $k = 13$ (1). | 1+1 |
| 14 | Zeroes $\alpha$ and $4\alpha$: $5\alpha = 15$ so $\alpha = 3$ (1); $k = 4\alpha^2 = 36$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\alpha+\beta = 7$, $\alpha\beta = 10$ (1); sum $= (\alpha+\beta) + \dfrac{\alpha+\beta}{\alpha\beta} = \frac{77}{10}$ (1); product $= \alpha\beta + 2 + \dfrac{1}{\alpha\beta} = \frac{121}{10}$ (1). | 1+1+1 |
| 16 | $(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $400 - 36 = 4c$ (1); $c = 91$ (1). | 1+1+1 |
| 17 | $\sum\alpha = 3$, $\sum\alpha\beta = -6$ (1+1); $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta = 21$ (1). | 1+1+1 |
| | **OR:** $\alpha+\beta = 7$, $\alpha\beta = 6$ (1); sum $= (\alpha+\beta) + \dfrac{\alpha+\beta}{\alpha\beta} = \frac{49}{6}$ (1); product $= \alpha\beta + 2 + \dfrac{1}{\alpha\beta} = \frac{49}{6}$ (1). | 1+1+1 |
| 18 | $(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $324 - 16 = 4c$ (1); $c = 77$ (1). | 1+1+1 |
| | **OR:** $\sum\alpha = 9$, $\sum\alpha\beta = 24$ (1+1); $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta = 33$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Let zeroes be $a-d, a, a+d$; sum $= 3a = 15$ so $a = 5$ (2); product $= a(a^2-d^2) = 120$ gives $d^2 = 1$ (2); zeroes $4, 5, 6$ (1). | 2+2+1 |
| | **OR:** $p(x) = g(x)q(x) + r(x)$ so $g(x) = \dfrac{p(x)-r(x)}{q(x)}$ (2); $p(x) - r(x) = g(x)q(x) = x^3 - 1$ (1); dividing, $g(x) = x - 1$ (2). | 2+1+2 |
| 20 | Let zeroes be $a-d, a, a+d$; sum $= 3a = 18$ so $a = 6$ (2); product $= a(a^2-d^2) = 120$ gives $d^2 = 16$ (2); zeroes $2, 6, 10$ (1). | 2+2+1 |
| | **OR:** $p(x) = g(x)q(x) + r(x)$ so $g(x) = \dfrac{p(x)-r(x)}{q(x)}$ (2); $p(x) - r(x) = g(x)q(x) = 2x^2 + 5x - 3$ (1); dividing, $g(x) = x + 3$ (2). | 2+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.